STAGE 1 — TECHNICAL ENGINEERING ASSESSMENT

TEAM ID	TEAM NAME

Problem Statement: Semiconductor Manufacturing Process Automations

A semiconductor manufacturing facility operates a number of critical-process automation and utility systems that supports wafer fabrication, chemical processing, slurry handling, cleaning and other high-precision manufacturing operations. These systems are required to operate continuously and reliably, with level of automation and minimum operator intervention. Process parameters such as flow, pressure, temperature, fluid properties, valve position and sensor signals must be accurately measured and controlled to maintain process stability and repeatability. Heat and flow parameters must be continuously monitored and accurately controlled to ensure uniform heat transfer, stable material flow and consistent process conditions throughout the manufacturing operation. In such environment's temperature and flow sensors, transmitters and control valves are used to detect deviations and automatically regulate the system, minimizing process variation and improving overall equipment reliability and product quality.

A typical industrial automation system consists of storage tanks, VFD pumps, transmitters, controllers, communication networks and safety interlocks. The system must maintain the required process conditions while responding appropriately to disturbances, equipment degradation and sensor failures.

In a semiconductor manufacturing, even relatively small deviations in process conditions can affect process repeatability, equipment performance, material utilization and overall manufacturing quality. Therefore, engineers must be able to analyse the physical process, select appropriate control and instrumentation methods, identify abnormal condition/graphs and develop reliable automated response.

Engineering Challenge

You are part of a multidisciplinary engineering team responsible for analysing and improving an automated semiconductor process handling system. The engineering team is expected to handle:

- Process hydraulics
- Valve selections
- Instrumentation and sensor calibration
- Closed-loop control
- Process safety and fault handling
- Industrial communication and EMI protection

The team must use the given process parameters and engineering information to calculate, analyse, justify and propose solutions for the problems presented in the following tasks.

Given Process Parameters

Pipe diameter	**10 mm**
Fluid density (ρ)	**1000 kg/m^3**
Dynamic viscosity (μ)	**1×10^{-3} Pa·s**
Required flow range	**1–6 L/min**
Maximum valve differential pressure (ΔP)	**1.5 bar**
Specific gravity (SG)	**1**
Pressure transmitter range	**0–4 bar**
ADC	**12 bit**
Sensor output	**4–20 mA**
Supply voltage	**24 VDC**
ΔT	**35°C**
Process pipeline length	**3 meters**
Darcy friction factor (f)	**0.03**
Plant transfer function	**G(s) = 6/(4s+1)**
Specific heat capacity (Cp)	**4.18 kJ/kg·K**
Heat-transfer coefficient (U)	**500 W/m^{2}K**
Gas constant for air (R)	**0.287 kPa·m^3/(kg·K)**
Atmospheric pressure	**1.013 bar**

Assume and state data where it is relevant.

TEAM INSTRUCTIONS

- **Print the Question paper and fill the answers by hand in blue/black ink only**
- Answer all questions clearly and completely **only within the space provided**. Show **only** the relevant formula, assumptions, calculation steps, units and final answer wherever applicable. **No additional sheets will be attached.**
- For conceptual questions, provide technically justified answers linked to the given process.
- Clearly state assumptions or missing data used.
- Use equations, sketches or diagrams where they improve clarity.
- Stay concise and within the space provided. Avoid overwriting in the given space.
- Submit original work, copying from other teams or external sources may result in disqualification.
- **Submission Guidelines:** 1. After answering all the questions **SCAN** and upload in PDF format only.
  2. One submission per team is only needed.
  3. Last date of submission is **13th SEPTEMBER 2026**, Time: **23:59Hr**

TASK 1 PIPE FLOW AND PROCESS HYDRAULICS

A process line transports fluid between storage and reaction sections. With required flow between 1 L/min and 6 L/min.

Q1.1 Calculate fluid velocity for: minimum flow & maximum flow (Hint: use formula) and also calculate the required heat-transfer rate for two flow conditions where ΔT = 35°C, assuming steady-state operation.

ANSWER

Q1.2 (a) Explain how change in fluid velocity and flow rate in a process pipeline can affect erosion, cavitation, pressure drop, mixing quality & equipment reliability. (b) Also explain how ΔT and heat transfer (U) influence heat-transfer rate, heating/cooling time, thermal efficiency, overheating risk and overall product quality. State engineering relationships using formulas to justify your answer.

ANSWER

Q1.3 Calculate Reynolds number for both min & max flow conditions (Hint: use formula). Classify it as laminar, transitional or turbulent & explain which is a better regime for precision dosing & why?

ANSWER

Q1.4 In the pipeline, estimate pressure drop over 3 m (Hint: use formula and take constant f = 0.03) and the effect of 20% scale deposition? Also, if the heating pipe has the length as mentioned, estimate the heat loss to the surroundings using a suitable heat transfer relation (use temperature difference 35°C).

ANSWER

TASK 2 VALVE ENGINEERING & FLOW CONTROL

A control valve regulates process flow. Maximum differential pressure: ΔP = 1.5 bar (take SG = 1).

Q2.1 A control valve is required to regulate a maximum process flow of 6 L/min at ΔP and SG (as above). Calculate the required Cv. A commercially selected valve has a rated Cv of at least 20% greater than the calculated requirement. Determine the selected Cv and calculate the resulting valve sizing margin. Explain why excessive oversizing of a control valve can adversely affect controllability at low flow.

ANSWER

Q2.2 Three valve options: ball valve, needle valve, globe valve. Select the most suitable and justify based on control linearity, maintenance and pressure drop. A valve hysteresis is observed to be 8% — explain the causes, detection methods & compensation strategy.

ANSWER

Q2.3 Air compressor delivers compressed air at 20–25 bar to two storage vessels using 2 SMC self-regulating valves. Vessel-1 holds air at 0–17 bar and vessel-2 holds air at 0–5.5 bar(g). A total of 3 process connections (A, B, C) are made from the two pressure vessels. Process line A is connected to vessel-2, where this line's temperature is 28°C and mass flow rate is 0.0055 kg/sec; this line contains 1 temperature transmitter, 1 pressure transmitter, 2 ON/OFF valves, 1 flow transmitter and 1 manual valve. For line A, calculate the volumetric flow in m^3/hr (given gas constant). Line B uses a SOV and provides pulsating pressure of 100 cycles at 0–15 bar, whereas line C acts as instrument air to the automatic ON/OFF valves mounted on process line A. Draw the P&ID diagram for all the process connections, pressure vessels and instrumentation.

P&ID DIAGRAM SPACE

Q2.4 A heat exchanger heats the process fluid from 25°C to 60°C using hot water entering at 80°C and leaving at 65°C. For the maximum flow rate of 6 L/min, calculate the required heat-transfer area using the LMTD method and (U) = 500 W/m^2K. Also briefly state one effect of 20% fouling on the required area.

ANSWER

TASK 3 SENSOR ENGINEERING AND CALIBRATION

Pressure measurement range: 0–4 bar, with sensor output 4–20 mA; temperature measurement range: 0–100°C; ADC: 12-bit.

Q3.1 Calculate pressure sensor resolution (write formula) and explain whether this is sufficient for ±1% control.

ANSWER

Q3.2 A temperature sensor has a range of 0–100°C and output 4–20 mA. Plot the calibration graph (temperature vs. current) and determine the temperature at 12 mA.

CALIBRATION GRAPH SPACE

Q3.3 Design a suitable 4–20 mA to 0–3.3 V signal-conditioning circuit. Calculate the required shunt resistor. Also explain why signal conditioning is required between the temperature transmitter and controller.

ANSWER

Q3.4 If the sensor drifts after 120 days to +0.08 bar, develop: calibration SOP, zero correction method, field validation procedure.

ANSWER

Q3.5 You have a ½-inch pipe with air pressure of 15 bar in it and an Aira Euro SOV connected to the pipe, running on 24 VDC with 8 W. The outlet pressure of this SOV is unstable and the coil gets heated up (SOV maximum rating is 20 bar, max temp 150°C). What should be done? (Answers can be structured as case-1, case-2, etc.)

ANSWER

TASK 4 CLOSED LOOP CONTROL ENGINEERING

Plant transfer function: G(s) = 6/(4s+1)

Q4.1 Determine: process gain, time constant and approximate setting time T(s) = 4τ.

ANSWER

Q4.2 A disturbance causes oscillation every 7 sec. Identify: possible causes, sensor delay effects and actuator saturation effects.

ANSWER

Q4.3 Design a simple hand sketch to transfer iron ore-slurry from a slurry tank (60% slurry and 40% water by weight) to a height of 15 meters from ground into a filter press machine placed horizontally on the 2nd floor (with 1 pump, 1 tank, 2 manual valves, 1 On/Off valve, 1 NRV, 2 pressure transmitters, 1 density meter, a VFD pump and a drain valve). Explain the formula and functions of the components of a PID controller.

HAND SKETCH SPACE

TASK 5 PROCESS SAFETY AND DIGITAL FAULT LOGIC

Q5.1 Sensor failure causes false low-level reading. Design fail-safe logic and draw the sequence.

FAIL-SAFE LOGIC / SEQUENCE SPACE

Q5.2 Sensor data is sampled at 100 Hz, total 12 sensors, with a runtime of 30 days. Write a pseudocode for anomaly detection: to detect spikes, missing packets and drift.

ANSWER

Q5.3 Draw a fused and unfused AI wiring for an Endress+Hauser vortex flow meter connected to an 8-channel analog input card on a 12-slot PLC rack. (Note: 24 VDC SMPS is part of the PLC panel.)

WIRING DIAGRAM SPACE

TASK 6 INDUSTRIAL COMMUNICATION

Q6.1 Temperature sensors are distributed across a 20 m process area where the environment contains motors, variable-frequency drives (VFDs), electrical heaters and welding machines. Choose between CAN, RS-485 and UART. Justify your selection based on noise immunity, communication distance, latency, scalability and reliability in an industrial thermal plant.

ANSWER

Q6.2 A pressure-monitoring packet contains 64 bytes and the communication bit rate is 115200 bps. (a) Calculate the approximate transmission time for one packet. (b) Assuming the serial communication uses 8N1 format (8 data bits, no parity, 1 stop bit), calculate the approximate transmission time required to send one complete packet.

ANSWER

Q6.3 A CPU installed in the PLC panel has an analog input conversion/sampling speed of 500 µs per channel. To acquire 2000 samples of a flow transmitter (FT) in 200 sec, for FT values ranging from approximately 2.2–3.5 m^3/hr, write a **logic sequence** for capturing the data and computing the mean mass flow rate (kg/s) (density = 5.204 kg/m^3), and **code 3-sigma deviation logic**. The overall CPU scan time for typical large program configurations runs around **15 ms**.

ANSWER

Q6.4 Explain the OSI model and describe the function of each of its seven layers. Compare the OSI model with the TCP/IP model.

ANSWER

Q6.5 What is an Industrial Internet of Things (IIoT) system? Explain its main components and how it is used in industries for real-time monitoring, data collection and process automation.

ANSWER